



PROJECT UPDATE REPORT

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CSE299 Section-3 | Group-1

Project Update Report

AI-Powered E-Commerce Platform with Intelligent Business Insights

1. Introduction

This report summarizes the development progress of second project update for our AI-Powered E-Commerce Platform with Intelligent Business Insights. The project aims to develop a modern e-commerce system that combines standard online shopping functionality with an AI-powered chatbot to improve customer interaction and provide intelligent assistance.

During this development phase, the team focused on three major components of the system: frontend development, backend implementation, and AI chatbot integration. These components were developed independently and then integrated to establish complete communication between the user interface, the server, and the AI service.

2. Frontend Development Progress

The frontend was developed using React with Vite to provide a responsive and user-friendly interface. Multiple pages were designed to support the overall shopping experience, including product browsing, shopping cart, checkout, and chatbot interaction.

Reusable components such as the navigation bar and page layouts were implemented to maintain consistency throughout the application. React Router was used for client-side navigation, allowing smooth transitions between pages without reloading the application.

The frontend was also prepared to communicate with backend REST API endpoints, enabling dynamic data retrieval and user interaction.

3. Backend Development Progress

The backend was developed using the Django framework to manage the application's business logic and database operations.

Core backend functionalities include user authentication, product management, shopping cart operations, order processing, and REST API development. Database models were created to store users, products, orders, and other related information. The backend provides API endpoints that allow the frontend to retrieve and update application data efficiently.

The backend architecture was designed to remain modular, making future feature additions easier while maintaining clean project organization.

4. AI Chatbot Integration

The primary objective of this update was successfully integrating the AI chatbot into the existing e-commerce platform.

The chatbot communication pipeline was implemented by connecting the React frontend with the Django backend, which then communicates with an external Large Language Model through an API. User prompts entered from the chatbot interface are transmitted to the backend, processed by the AI model, and the generated responses are returned to the frontend for display.

This integration establishes complete end-to-end communication between the frontend interface, backend services, and the AI model. Initial testing confirmed successful prompt submission, response generation, and proper rendering of chatbot replies within the application.

5. Current Project Status

At the completion of this update, the project contains a functional frontend interface, a working Django backend, and an integrated AI chatbot capable of responding to user queries through natural language interaction.

The individual development efforts of each team member have been combined successfully into a single working application. Internal testing demonstrates that the system components communicate correctly and perform their intended functions.

6. Future Work

The next development phase will focus on enhancing chatbot capabilities, improving user experience, expanding backend functionality, and performing additional testing. Future work will also include optimizing application performance, strengthening error handling, and integrating the remaining planned features described in the project proposal.